

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – IV

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Date of Assignment: 20/08/2026	Date/Time of Submission: 20/08/2026

Aim: Performing Data Analysis using Calculations in Tableau

1. Create basic calculated fields using formulas

Step 1: Connect the Dataset

1. Open Tableau Desktop.
2. Under Connect, click Text File.
3. Select your dataset ([tableau_dataset.csv](#)).
4. Click Open.
5. Click Sheet 1 to begin analysis.

Step 2: Create the Calculated Field

1. In the Data Pane, right-click anywhere in the blank area.
2. Select Create → Calculated Field.
3. Name the calculated field Profit Margin.
4. Enter the following formula:

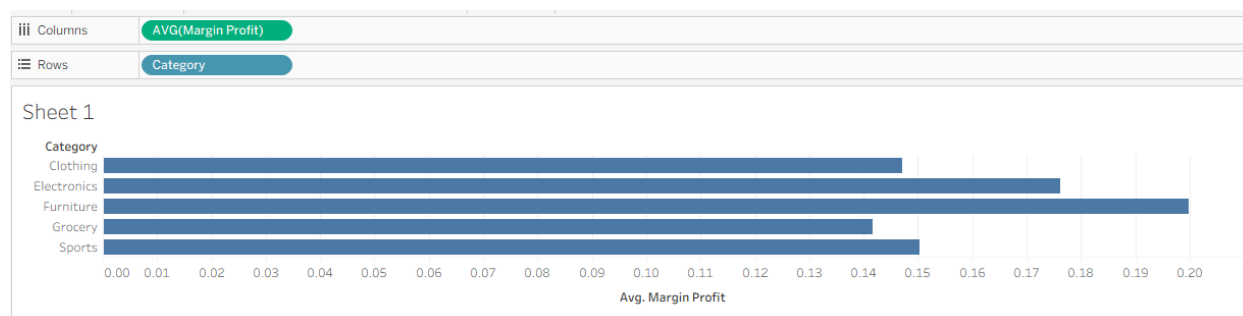
$$[\text{Profit}] / [\text{Sales}]$$
5. Verify that the message "The calculation is valid" appears.
6. Click OK.

Step 3: Use the Calculated Field

1. Drag Category to the Rows shelf.
2. Drag Profit Margin to the Columns shelf.
3. Right-click the Profit Margin pill.
4. Select Measure → Average.

Why We Are Doing It

A basic calculated field creates a new measure using mathematical operations. Here, the Profit Margin indicates the percentage of profit earned from total sales.



2. Perform row-level calculations

Step 1: Create the Calculated Field

1. Right-click in the Data Pane.
2. Select Create → Calculated Field.
3. Name it Revenue Before Discount.
4. Enter the formula:

[Sales] * [Quantity]

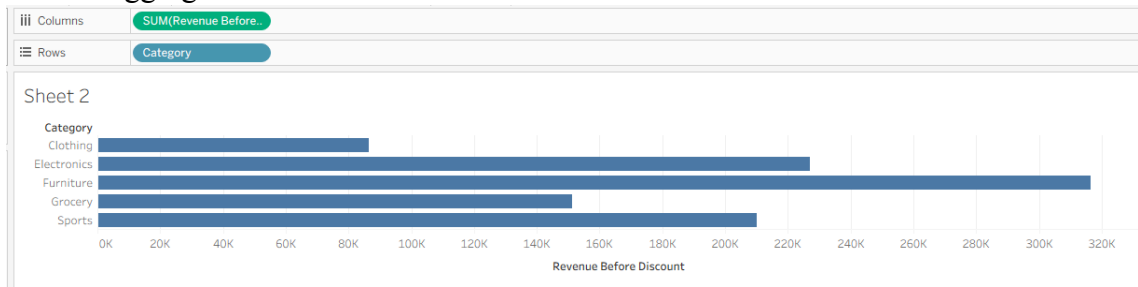
5. Ensure the calculation is valid.
6. Click OK.

Step 2: Display the Result

1. Drag Category to the Rows shelf.
2. Drag Revenue Before Discount to the Columns shelf.
3. Compare it with the Sales measure if required.

Why We Are Doing It

Row-level calculations perform mathematical operations on every individual record before Tableau aggregates the data.



3. Perform aggregate-level calculations

Step 1: Create the Calculated Field

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name it Aggregate Profit Margin.
4. Enter the formula:

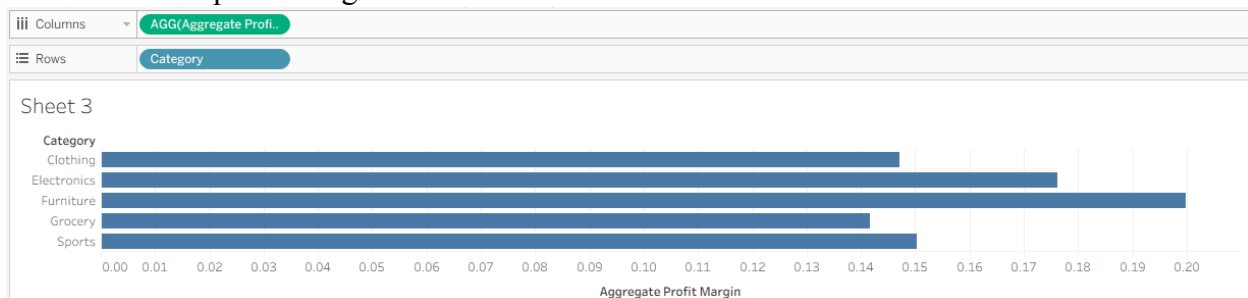
$$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$$
5. Click OK.

Step 2: Display the Result

1. Drag Category to the Rows shelf.
2. Drag Aggregate Profit Margin to the Columns shelf.

Why We Are Doing It

Aggregate calculations first summarize the data and then perform the calculation, providing an accurate overall profit margin.



4. Implement FIXED Level of Detail (LOD) calculations

Step 1: Create the Calculated Field

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name it Fixed State Sales.
4. Enter the formula:

$$\{ \text{FIXED} [\text{State}] : \text{SUM}([\text{Sales}]) \}$$
5. Click OK.

Step 2: Display the Result

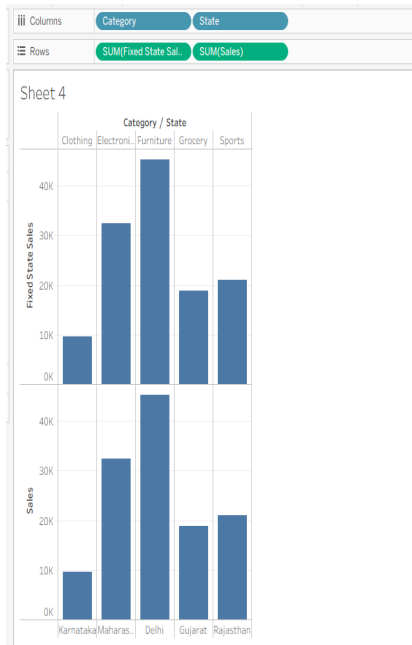
1. Drag State to the Rows shelf.
2. Drag Category beside State.
3. Drag Sales to the Columns shelf.
4. Drag Fixed State Sales next to Sales.

Observation

The Sales values change according to each category, whereas Fixed State Sales remains constant for each state.

Why We Are Doing It

A FIXED LOD calculation computes a value at a specified level of detail regardless of the dimensions displayed in the worksheet.



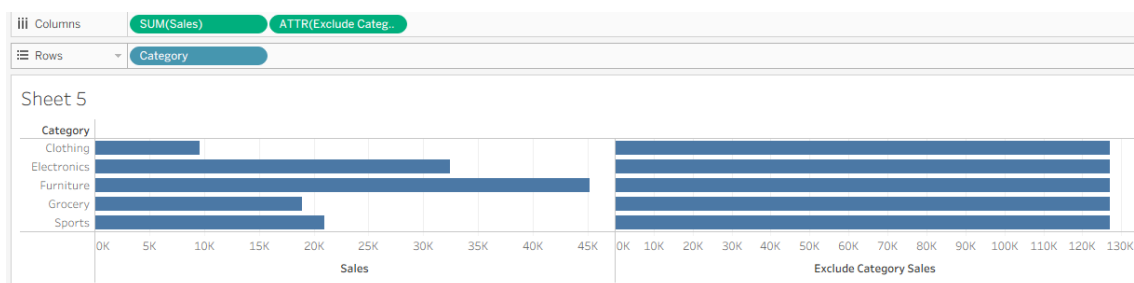
5. Use INCLUDE and EXCLUDE LOD expressions

Steps

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name it Exclude Category Sales.
4. Enter:
`{ EXCLUDE [Category] : SUM([Sales]) }`
5. Click OK.
6. Drag Category to the Rows shelf.
7. Drag Sales to the Columns shelf.
8. Drag Exclude Category Sales to the Columns shelf beside Sales.

Why We Are Doing It

The EXCLUDE LOD expression ignores the Category field while calculating total sales, allowing comparison between category sales and overall sales.



6. Create and use parameters for dynamic analysis

Step 1: Create the Parameter

1. Right-click in the Data Pane.
2. Select Create Parameter.
3. Name the parameter Profit Target.
4. Set:
 - Data Type: Integer
 - Current Value: 10000
5. Click OK.

Step 2: Create the Calculated Field

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name it KPI Threshold Met.
4. Enter:

```
IF SUM([Profit]) >= [Profit Target] THEN
"Target Achieved"
ELSE
"Below Target"
END
```
5. Click OK.

Step 3: Display the Result

1. Drag State to the Rows shelf.
2. Drag Profit to the Columns shelf.
3. Drag KPI Threshold Met to the Color option in the Marks card.
4. Right-click Profit Target and select Show Parameter.
5. Change the parameter value to observe how the chart updates dynamically.

Why We Are Doing It

Parameters allow users to interactively change calculation thresholds without modifying the original data.

Create Parameter

Name
Profit Target

Properties

Data type
Integer

Display format
10,000

Current value
10,000

Value when workbook opens
Current value

Allowable values
☒ All ☐ List ☐ Range

Cancel OK

7. Apply calculations to fix data issues and enhance insights

Steps

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name it High Profit Category.
4. Enter:

```
IF [Profit] > 5000 THEN
```

"High Profit"

ELSE

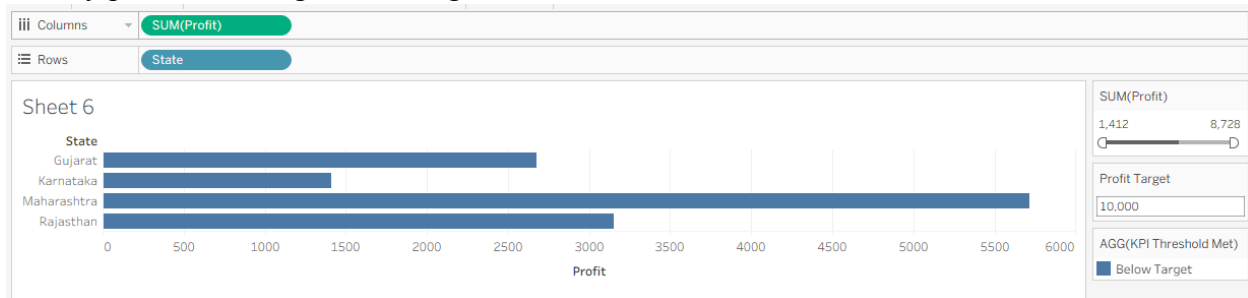
"Low Profit"

END

5. Click OK.
6. Drag Category to the Rows shelf.
7. Drag Sales to the Columns shelf.
8. Drag High Profit Category to the Color option in the Marks card.

Why We Are Doing It

This calculation classifies records into High Profit and Low Profit groups, making it easier to identify profitable categories through color-based visualization.



Conclusion:-

In this practical, Tableau's calculated fields were used to perform basic, row-level, and aggregate calculations. FIXED and EXCLUDE LOD expressions were implemented to analyze data at different levels of detail, and parameters enabled dynamic interaction with the visualization. Finally, a custom calculated field was created to classify profit levels, demonstrating how calculations can improve business insights and support effective decision-making.